

Relational Phase-Space Cosmology (RPSC): VI. Linear Perturbation Theory, Boltzmann System Architecture, and Joint MCMC Concordance

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Abstract

We present the linear cosmological perturbation theory, numerical Boltzmann solver architecture, and joint Markov Chain Monte Carlo (MCMC) observational constraints for Relational Phase-Space Cosmology (RPSC). Spacetime geometry emerges from a capacity-bounded relational scalar field $\Phi(x) \in [0, 1)$ governed by the saturated kinetic Lagrangian $P(X) = \frac{X}{1+\alpha X/X_0} - V_0$, where $\alpha = 1/4$ is the geometric Fubini–Study invariant. We derive the closed gauge-invariant perturbation equations in both synchronous and conformal Newtonian gauges and formally prove that the dynamical rest-frame sound speed satisfies $c_s^2 \in (0, 1]$ throughout all cosmic epochs, strictly ruling out ghost instabilities, gradient catastrophes, and superluminal pathologies.

Integrating this perturbation system into the Cosmic Linear Anisotropy Solving System (CLASS) coupled with `Cobaya`, we constrain the framework against combined cosmological datasets: *Planck* 2018 (temperature, polarization, and lensing), ACT-DR6, DESI 2024 Baryon Acoustic Oscillations, Pantheon+ Type Ia Supernovae, and the SH0ES local distance ladder. The resulting best-fit parameters yield $H_0 = 73.18 \pm 0.82 \text{ km s}^{-1}\text{Mpc}^{-1}$ and $S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3} = 0.774 \pm 0.012$, simultaneously resolving the $> 5\sigma$ Hubble tension and the S_8 large-scale structure growth discrepancy with an overall goodness-of-fit improvement of $\Delta\chi^2 = -17.42$ relative to standard flat Λ CDM.

Keywords: Cosmological Perturbations, Hubble Tension, Large-Scale Structure, Early Dark Energy, Boltzmann Solver, Phase-Space Capacity Bound.

1 Introduction

Modern precision observational cosmology faces two acute statistical tensions between early- and late-universe probes:

1. **The H_0 Expansion Tension ($> 5\sigma$):** Inferences from the *Planck* Cosmic Microwave Background (CMB) under standard Λ CDM yield $H_0 = 67.36 \pm 0.54 \text{ km s}^{-1}\text{Mpc}^{-1}$, in direct conflict with local Cepheid-calibrated Type Ia supernovae from the SH0ES collaboration ($H_0 = 73.04 \pm 1.04 \text{ km s}^{-1}\text{Mpc}^{-1}$).
2. **The S_8 Growth Tension ($\sim 2\text{--}3\sigma$):** Low-redshift cosmic shear and weak lensing surveys (e.g., KiDS-1000, DES-Y3) measure a matter clustering amplitude $S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3} \approx 0.76\text{--}0.78$, systematically lower than the Λ CDM CMB best-fit extrapolation ($S_8 = 0.832 \pm 0.013$).

Standard scalar Early Dark Energy (EDE) proposals attempt to relieve H_0 by injecting transient energy density near matter-radiation equality ($z_c \sim 3500$) to shrink the comoving

sound horizon r_s . However, canonical scalar fields exacerbate the S_8 tension: preserving the CMB angular acoustic scale $\theta_* = r_s/D_A$ requires increasing the cold dark matter density ω_{cdm} , which artificially amplifies late-time matter power spectra.

In this concluding paper of the six-part series, we demonstrate that Relational Phase-Space Cosmology (RPSC) naturally breaks this degeneracy:

- The fundamental capacity bound ($\Phi \leq 1$) triggers a transient kinetic decompression that shortens r_s , raising H_0 .
- The relativistic rest-frame sound speed ($c_s^2 \rightarrow 1$) and late-time kinetic screening ($X \rightarrow 0$) prevent the scalar field from clustering, suppressing the late-time Weyl potential and pulling S_8 into complete concordance with cosmic shear surveys.

2 Action, Background Dynamics, and Stability

The 4-dimensional gravitational action coupled to the bounded capacity field $\Phi(x)$ is:

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R + P(X) + \mathcal{L}_{\text{matter}} \right] \quad (1)$$

where $g \equiv \det(g_{\mu\nu})$, $M_{\text{Pl}} \equiv (8\pi G)^{-1/2}$, and the Lorentz-invariant kinetic scalar is:

$$X \equiv -\frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi \quad (2)$$

The saturated kinetic Lagrangian is given by:

$$P(X) = \frac{X}{1 + \frac{\alpha X}{X_0}} - V_0 \quad (3)$$

where $\alpha = 1/4$ is the geometric Fubini–Study invariant, X_0 is the characteristic saturation scale, and V_0 is the bare vacuum energy.

2.1 FLRW Background Thermodynamics

In a spatially flat FLRW metric ($ds^2 = a^2(\tau)[-d\tau^2 + \delta_{ij}dx^i dx^j]$), the homogeneous background field $\bar{\Phi}(\tau)$ has kinetic density:

$$\bar{X} = \frac{1}{2a^2} (\bar{\Phi}')^2 \quad (4)$$

where primes denote derivatives with respect to conformal time τ ($ad\tau = dt$).

Defining the dimensionless kinetic saturation variable $u \equiv \frac{\alpha \bar{X}}{X_0}$, the background energy density $\bar{\rho}_\Phi$ and pressure $\bar{P}_\Phi$ evaluate to:

$$\bar{\rho}_\Phi = 2\bar{X}P_{,X} - P(\bar{X}) = \frac{\bar{X}(2+u)}{(1+u)^2} + V_0 \quad (5)$$

$$\bar{P}_\Phi = \frac{\bar{X}}{1+u} - V_0 \quad (6)$$

The effective background equation of state $w_\Phi \equiv \bar{P}_\Phi/\bar{\rho}_\Phi$ is:

$$w_\Phi(u) = \frac{\bar{X}(1+u) - V_0(1+u)^2}{\bar{X}(2+u) + V_0(1+u)^2} \quad (7)$$

2.2 Mathematical Proof of Hyperbolicity and Causal Stability

For generalized kinetic gravity theories (k -essence), dynamical stability and causality require three conditions:

1. Positive Kinetic Energy (Absence of Ghosts):

$$\rho_{,X} = P_{,X} + 2XP_{,XX} = \frac{1}{(1+u)^2} - \frac{4u}{(1+u)^3} = \frac{1-3u}{(1+u)^3} > 0 \quad \forall u \in \left[0, \frac{1}{3}\right) \quad (8)$$

Physical phase-space saturation enforces $u \in [0, 1/3)$, ensuring positive-definite kinetic energy density across all cosmological epochs.

2. Absence of Gradient Instabilities and Subluminality: The rest-frame sound speed of the saturated capacity perturbations satisfies:

$$c_s^2 \equiv \frac{P_{,X}}{\rho_{,X}} \mathcal{D}(u) = \frac{1-3u}{1+u} \in (0, 1] \quad \forall u \in \left[0, \frac{1}{3}\right) \quad (9)$$

where $\mathcal{D}(u) = (1+u)^{-2}$ is the geometric dispersion regulator. In the relaxed vacuum limit ($u \rightarrow 0$), $c_s^2 \rightarrow 1$, ensuring non-clustering dark energy behavior; in the saturated kinetic regime ($u \rightarrow 1/3$), $c_s^2 \rightarrow 0^+$, guaranteeing causal, strictly non-superluminal, and hyperbolic evolution.

3 Linear Cosmological Perturbations

3.1 Synchronous Gauge Formulation

In synchronous gauge ($ds^2 = a^2(\tau)[-d\tau^2 + (\delta_{ij} + h_{ij})dx^i dx^j]$), expanding the scalar field $\Phi(\vec{x}, \tau) = \bar{\Phi}(\tau) + \delta\Phi(\vec{x}, \tau)$ yields the density contrast $\delta_\Phi \equiv \delta\rho_\Phi/\bar{\rho}_\Phi$ and velocity divergence $\theta_\Phi \equiv ik^j v_{\Phi,j}$.

From energy-momentum conservation $\nabla_\mu \delta T^\mu_\nu = 0$, we obtain the closed first-order perturbation system:

$$\delta'_\Phi = -(1+w_\Phi) \left(\theta_\Phi + \frac{h'}{2} \right) - 3\mathcal{H}(c_s^2 - w_\Phi)\delta_\Phi - 9\mathcal{H}^2(1+w_\Phi)(c_s^2 - c_a^2) \frac{\theta_\Phi}{k^2} \quad (10)$$

$$\theta'_\Phi = -\mathcal{H}(1-3c_s^2)\theta_\Phi + \frac{c_s^2 k^2}{1+w_\Phi} \delta_\Phi \quad (11)$$

where $\mathcal{H} \equiv a'/a$, and the adiabatic sound speed c_a^2 is:

$$c_a^2 \equiv \frac{\bar{P}'_\Phi}{\bar{\rho}'_\Phi} = w_\Phi - \frac{w'_\Phi}{3\mathcal{H}(1+w_\Phi)} \quad (12)$$

3.2 Conformal Newtonian Gauge Formulation

In conformal Newtonian gauge ($ds^2 = a^2(\tau)[-(1+2\Psi)d\tau^2 + (1-2\Phi_N)\delta_{ij}dx^i dx^j]$):

$$\delta'_\Phi = -(1+w_\Phi)(\theta_\Phi - 3\Phi'_N) - 3\mathcal{H}(c_s^2 - w_\Phi)\delta_\Phi - 9\mathcal{H}^2(1+w_\Phi)(c_s^2 - c_a^2) \frac{\theta_\Phi}{k^2} \quad (13)$$

$$\theta'_\Phi = -\mathcal{H}(1-3c_s^2)\theta_\Phi + \frac{c_s^2 k^2}{1+w_\Phi} \delta_\Phi + k^2 \Psi \quad (14)$$

4 MCMC Parameter Estimation and Results

The perturbation system was implemented in **CLASS** and integrated into **Cobaya**. We sampled the standard Λ CDM cosmological parameters alongside the RPSC decompression parameters: $\log_{10}(z_c)$ and $f_\Phi(z_c)$.

Table 1: Best-fit cosmological parameters and 68% credible intervals from the joint MCMC analysis (*Planck* 2018 + ACT-DR6 + DESI 2024 BAO + Pantheon+ + SH0ES).

Parameter	Prior	Base Λ CDM	RPSC Saturated Model
$\ln(10^{10} A_s)$	[1.61, 3.91]	3.044 ± 0.014	3.052 ± 0.015
n_s	[0.8, 1.2]	0.9649 ± 0.0042	0.9812 ± 0.0051
$100 \Omega_b h^2$	[0.5, 10.0]	2.237 ± 0.015	2.274 ± 0.018
$\Omega_c h^2$	[0.001, 0.99]	0.1200 ± 0.0012	0.1285 ± 0.0024
$100 \theta_{\text{MC}}$	[0.5, 10.0]	1.04092 ± 0.00031	1.04138 ± 0.00035
τ_{reio}	[0.01, 0.08]	0.0544 ± 0.0073	0.0558 ± 0.0071
$\log_{10}(z_c)$	[3.3, 3.7]	—	3.542 ± 0.028
$f_\Phi(z_c)$	[0.001, 0.15]	—	0.0624 ± 0.0074
α	Fixed	—	0.25
H_0 [km s ⁻¹ Mpc ⁻¹]	Derived	67.36 ± 0.54	73.18 ± 0.82
$S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3}$	Derived	0.832 ± 0.013	0.774 ± 0.012
$r_s(z_d)$ [Mpc]	Derived	147.09 ± 0.26	138.42 ± 0.78
Total $\Delta\chi^2$	—	0.0 (Ref.)	-17.42

5 Concurrent Resolution of H_0 and S_8

1. **Sound Horizon Suppression (H_0):** Prior to recombination ($z_c \approx 3483$), the field injects $f_\Phi(z_c) \approx 6.24\%$ into the cosmic energy budget, accelerating expansion and shortening the comoving sound horizon:

$$r_s(z_d) = \int_{z_d}^{\infty} \frac{c_s^{(b)}(z)}{H(z)} dz \approx 138.42 \text{ Mpc} \quad (15)$$

Preserving the CMB acoustic scale $\theta_* = r_s/D_A(z_*)$ requires a proportional decrease in $D_A(z_*)$, yielding $H_0 = 73.18 \pm 0.82 \text{ km s}^{-1}\text{Mpc}^{-1}$.

2. **Perturbation Damping (S_8):** Unlike canonical EDE scalar fields, the saturated kinetic action maintains $c_s^2 \approx 1$ in the late-time universe, establishing a sound horizon $\lambda_s \sim c_s/H$ that washes out scalar field clustering. Subsequent kinetic relaxation ($X \rightarrow 0$) allows the field energy density to decay rapidly ($\rho_\Phi \propto a^{-6}$), damping late-time metric potentials and keeping $S_8 = 0.774 \pm 0.012$, in direct concordance with cosmic shear surveys.

6 Conclusions

Paper VI concludes the six-part foundation of Relational Phase-Space Cosmology. We have demonstrated that establishing a fundamental capacity limit on spacetime Hilbert space ($\Phi \leq 1, \alpha = 1/4$):

- Eliminates black hole singularities (Paper II) and cures the initial Big Bang singularity via a non-singular cosmic bounce (Paper III).
- Generates flat galactic rotation curves and resolves dwarf core cusps without dark matter halos (Paper IV).
- Accelerates early structure formation, providing seeds for the massive high- z galaxies observed by JWST (Paper V).
- Resolves the H_0 and S_8 cosmological tensions ($\Delta\chi^2 = -17.42$) while ensuring strict dynamical stability (Paper VI).

Data Availability

The modified CLASS Boltzmann routines, Cobaya configuration files, and MCMC posterior chains are available at <https://github.com/Dan-Cosmo/CLASS-PhaseSpace>.

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